Urea

Urea

- Its formula is H_2NCONH_2.
- Urea is the chief nitrogenous end product of the metabolic breakdown of proteins in all mammals and some fishes.
- The material occurs not only in the urine of all mammals but also in their blood, bile, milk, and perspiration.

Metabolism

- In the course of the breakdown of proteins, amino groups (NH_2) are removed from the amino acids that partly comprise proteins.
- These amino groups are converted to ammonia (NH_3), which is toxic to the body and thus must be converted to urea by the liver.
- The urea then passes to the kidneys and is eventually excreted in the urine.

Clinical importance

- The measurement of urea is an important investigation in diagnosing kidney damage.
- An adult typically excretes about 25 grams of urea per day.
- The normal range of blood urea is 15–40 mg/dl.
- Any condition which impairs the elimination of urea by the kidneys can lead to uremia, a buildup of urea and other nitrogen wastes in the blood.

Estimation of blood urea

- Hench - Aldrich method.
- Di-acetylmonoxime method.
- Urease and Berthelot reaction.

Hench - Aldrich method

- The degree of urea retention can be estimated clinically by the simple measurement of the mercury combining power of the deproteinized blood.

Di-acetylmonoxime method

- This protocol uses diacetyl monoxime to directly determine the concentration of urea in solution.
- The reaction between diacetyl monoxime and urea in the presence of sulfuric acid, phosphoric acid, thiosemicarbazide, and ferric chloride produces a chromophore with a peak absorbance at 520 nm.
- A linear relationship between concentration and absorbance from 0.4 to 5.0 mM urea can be analysed in this protocol.

Principle

- Diacetyl monoxime breaks down into diacetyl during the reaction in the presence of heat (provided from a boiling water bath).
- Diacetyl and urea then condense in the presence of a strong acid to form the yellow-colored diazine product and water.
- Diazine is stabilized by thiosemicarbazide and converted to a pinkish red-colored complex with a stronger absorbance in the presence of ferric ions derived from ferric chloride hexahydrate.
- The mixed acid reagent is stable for at least a month at room temperature.
- The mixed color reagent is stable for at least a week.
- The maximum absorption of the final product is at 520 nm and is proportional to the concentration of urea.

Urease and Berthelot reaction

- Berthelot's reagent is an alkaline solution of phenol and hypochlorite.
- Ammonia reacts with Berthelot's reagent to form a blue product which is used in a colorimetric method for determining ammonia.
- The reagent can also be used for determining urea.

Creatinine

- Creatinine is a waste product made by muscles as part of regular, everyday activity.
- Normally, kidneys filter creatinine from blood and send it out of the body in urine.
- If there is a problem with kidneys, creatinine can build up in the blood.